

NeCTv2: Optimizing Implicit Neural Representations for Discrete and Continuous 4D-CT

NeCT QuadCubes

Part I — Encoder Optimisation

QuadCubes baseline

44.8 GB · 35.89 dB

−45.8% VRAM

CombinedCubes

24.3 GB · 37.18 dB

+1.84 dB vs baseline

MixedCubes ★

22.1 GB · 37.73 dB

Part II — Continuous Scanning



Step-and-shoot

discrete angle acquisition

*corrects angular
integration blur*



Fly-scan + K-step correction

continuous-angle acquisition

Faster reconstructions · Lower hardware requirements · New acquisition mode for 4D-CT